WEEK 1 · SESSION 2

Enterprise Data Engineering, Analytics & AI

Data Flow in Applications

INPUT ▶ PROCESS ▶ STORAGE ▶ OUTPUT

How data moves from user action to business intelligence

Duration: 1 Hour | Trainer: Prachi Kabra

Session Compass

What this session covers

OBJECTIVE

1. Understand how data flows from user action to storage
2. Map the Input → Process → Storage → Output model
3. Identify how APIs and events carry data between systems
4. Recognise databases, APIs, dashboards in a real app
5. Trace data flow in 3 industry use cases end-to-end

Every App Has a Hidden Journey

Behind every tap, click and swipe — data is moving through a pipeline

1 INPUT

User triggers an event

- Form submit
- Button click
- Sensor ping

→

2 PROCESS

App logic transforms data

- Validate
- Calculate
- Transform

→

3 STORAGE

Database or cloud stores it

- SQL Table
- File / Cache
- Cloud store

→

4 OUTPUT

Dashboard, API or notification

- Report
- API response
- Alert / UI

💬 *'When you order food on Swiggy, what do you think happens in the first 0.5 seconds?'*

STAGE 1 OF 4 — INPUT

The Input Layer

Every data journey begins with a human (or machine) taking an action

👤 Human Input

- Typing in a form field
- Clicking 'Place Order'
- Uploading a document
- Speaking to a voice assistant
- Scanning a QR code

⚙️ System / App Input

- Scheduled batch jobs
- API calls from other apps
- Webhook triggers
- Background sync events
- Automated scripts / bots

📡 Machine / IoT Input

- Temperature sensors
- GPS location pings
- RFID scanner reads
- Camera / image capture
- Payment terminal swipes

🔑 **Key Insight: Every input must be captured, validated and formatted BEFORE it enters storage. Garbage in = Garbage out.**

STAGE 2 OF 4 — PROCESS

The Processing Layer

What 'Processing' actually means:

Validate: Check: is the email valid? Is the price positive?

Transform: Convert formats: date strings → timestamps, USD → INR

Enrich: Add context: customer segment, geo from IP, risk score

Route: Decide: which database? Which service? Which queue?

Compute: Run calculations: totals, discounts, recommendations

🍕 Real Example: Food Delivery App

1. User clicks 'Order Now' → Input received
2. App validates: item in stock? → Validate
3. Calculates total + delivery fee → Compute
4. Checks distance, assigns driver → Enrich
5. Sends to restaurant's system → Route
6. Saves order record to DB → Store

💬 *Activity: Trace what 'processing' a hospital patient registration triggers.*

The Storage Layer

Not all data lands in the same place — where it goes depends on HOW it will be used

Relational DB

MySQL, SQL Server

Transactional data

Data Warehouse

Snowflake, BigQuery

Analytics & reporting

Data Lake

Azure ADLS, S3

Raw / big data store

Cache / Redis

Redis, Memcached

Fast, temporary data

Cloud / File

Blob, S3, GCS

Documents, media

🔑 **How fast does this data need to be accessed? How long must it be kept?**

STAGE 4 OF 4 — OUTPUT

The Output Layer

Data only creates value when it reaches someone — in the right form, at the right time

📊 Dashboards & Reports

BI tools (Power BI, Tableau) pull data from the warehouse and present KPIs, trends, charts to business users

Consumer: Managers, Executives

🔌 API Responses

Backend APIs serve data to front-end apps, mobile apps, or other systems — in real time or near-real time

Consumer: Developers, Apps

🔔 Alerts & Notifications

Triggered outputs: emails, push notifications, SMS, Slack messages — when data meets a condition or threshold

Consumer: End Users, Operators

🤖 AI / ML Predictions

Models consume stored data and return predictions, recommendations, or classifications as output

Consumer: Products, Analysts

💬 *Which type of output does you use most?*

APIs — The Connective Tissue of Data Flow

APIs are what allow data to move between separate systems without direct database access

API (Application Programming Interface) = A contract that says: **'Send me THIS → I'll give you THAT'**

How an API call works:

1. Client sends REQUEST App → API endpoint with data payload

2. Server PROCESSES Validates auth, queries database, runs logic

3. Server sends RESPONSE Returns JSON/XML with result or error

4. Client RENDERS output App displays result to user

```
// API Response (JSON)

{
  "order_id": "ORD-9821",
  "status": "confirmed",
  "customer": {
    "name": "Ravi Shah",
    "city": "Surat"
  },
  "total": 450.00,
  "estimated_mins": 28
}
```

💡 *Every app you use daily — Swiggy, Google Maps, your bank — exchanges data via APIs exactly like this.*

⚡ QUIZ BREAK

3 Questions · 4 Minutes · Eyes on your own screen!

Q1 → Name the 4 stages of the data flow pipeline we covered today.

(Think: what happens before the database)

Q2 → What is the role of an API in a data flow?

(Hint: it connects two separate systems)

Q3 → A user clicks 'Pay Now' on an e-commerce site. List 3 things that happen BEFORE the payment record is saved.

(Think: validate, compute, route…)

(Answers revealed on next slide — no peeking!)

Data Flow — 3 Industries, End to End

Same 4-stage model — different technology stack every time

Banking — Loan Application

INPUT	**PROCESS**	**STORAGE**	**OUTPUT**
Customer submits form online	Credit score check, risk model runs	Application saved to core banking DB	Approval SMS + dashboard update

Healthcare — Patient Check-In

INPUT	**PROCESS**	**STORAGE**	**OUTPUT**
Staff scans patient ID card	Matches record, checks allergies	EHR updated in hospital system	Doctor receives alert on tablet

E-Commerce — Product Search

INPUT	**PROCESS**	**STORAGE**	**OUTPUT**
User types 'red shoes size 9'	Search engine ranks by relevance	Search query logged for analytics	Results + personalised ads shown

challenge: Pick an industry from YOUR experience — trace the data flow for one key action.

The Full Picture — Modern App Data Architecture

How all the layers connect in a real enterprise application

PRESENTATION LAYER	Web App | Mobile App | Dashboard | Chatbot
API / GATEWAY	REST API | GraphQL | Event Bus | Message Queue
BUSINESS LOGIC	Validation | Rules Engine | Calculation | Routing
DATA STORAGE	SQL DB | Data Lake | Cache (Redis) | File Store
ANALYTICS OUTPUT	BI Reports | ML Model Scoring | Alerts | Exports

↔ Data flows UP and DOWN — a dashboard query flows top→bottom to DB; a user form flows bottom→top as stored data becomes output

DATA STORAGE PATTERN

In the data pipeline: Bronze = Storage (raw), Silver = Storage (processed), Gold = Output-ready — all three layers live inside your Data Lake / Lakehouse (e.g. Databricks Delta Lake)

Medallion Architecture (modern data architecture)

A layered data design pattern that progressively refines raw data into trusted, analytics-ready information

🥉 BRONZE

Raw / Landing Zone

Data lands here exactly as it arrives — no transformation, no cleaning. This is your safety net: if anything goes wrong downstream, you always have the original.

What lives here:

- Raw API payloads (JSON/XML)
- CSV files from partners
- Event logs from apps
- IoT sensor dumps

Never delete. Never modify. Append only.

🥈 SILVER

Cleansed / Conformed

Bronze data is cleaned, deduplicated, validated and standardised. Columns are renamed, data types enforced, nulls handled. This layer is fully trusted.

What lives here:

- Cleaned customer records
- Standardised date formats
- Deduplicated transactions
- Validated & typed fields

Data here is consistent, correct and reusable.

🥇 GOLD

Business-Ready / Aggregated

Silver data is aggregated, enriched and modelled for business consumption. Fact tables, dimension tables, KPIs and metrics live here. This is what Power BI connects to.

What lives here:

- Monthly revenue summaries
- Customer 360 profiles
- KPI metric tables
- Star / Snowflake schema tables

Optimised for analytics. Fast. Trusted. Governed.

Session Summary

1. Every application follows the same core flow: INPUT → PROCESS → STORAGE → OUTPUT
2. Inputs come from humans, systems, and machines (IoT sensors, APIs, bots)
3. Processing transforms, validates, enriches and routes data before it is stored
4. Storage type depends on access speed and retention needs — DB vs Warehouse vs Lake vs Cache
5. APIs are the connective tissue — they allow data to flow between separate systems securely
6. Output forms include dashboards, API responses, alerts and AI predictions
7. Modern data Architecture

Questions?

"Every time you use an app — now you see the flow."

Activity

Pick one app you use daily.
Trace its data flow: What is the Input? What Processing must happen? Where is it Stored? What Output do YOU see?